

Progesterone and Negative Emotionality Across and Between Ovulatory Cycles:
A Study of Romantically Involved Women

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Abstract

In prior work, naturally cycling women's progesterone levels were found to be associated with their anxiety levels and concerns about levels of social support. The current study further examined these associations. Naturally cycling partnered women ($N = 180$) participated in up to four in-lab sessions across a month. Each session, they filled out measures of their mood states and their concerns about investment by their primary relationship partners. Progesterone, estradiol, testosterone, and cortisol levels were assayed in session-specific urine samples. In mixed model analyses, both average levels of hormones and within-woman variations of hormones were entered as predictors of negative emotionality and concerns about partner investment. Women's mean levels of progesterone positively predicted these outcomes, but their session-specific variations in progesterone were weakly and not significantly associated with these outcomes. Mean estradiol levels negatively and mean testosterone levels positively predicted negative emotionality. Only within-woman variations in cortisol levels predicted negative emotionality. In additional analyses, both mean follicular and luteal phase levels of progesterone predicted negative emotionality. Overall, results are not consistent with progesterone affecting negative emotionality. Perhaps negative emotionality influences progesterone levels, though additional research is needed before definitive causal conclusions can be offered.

KEYWORDS: progesterone; estradiol; ovarian hormones; negative affect; social support

1. Introduction

In industrialized populations, the average woman ovulates roughly 300 times in her lifetime. In traditional natural fertility populations, women may ovulate fewer than 100 times in a lifetime on average; they are pregnant or breastfeeding during much of their reproductive years (Strassmann, 1999). Only during a few days leading to ovulation can copulation result in conception and, therefore, pregnancy (e.g., Dunson et al., 2001). In total, these days amount to a small proportion of the average lifespan of a woman in a natural fertility population. Every human ever born was conceived during these very brief windows, successfully implanted in the uterine endometrium, and survived gestation.

Not surprisingly, then, natural selection has shaped adaptations that foster successful ovulation, conception, implantation, and gestation. The reproductive cycle entails substantial physiological changes, the timing of which are orchestrated by ovarian hormones, estrogens and progesterone. Psychological and behavioral adjustments likely accompany these physiological changes, further enhancing the prospects of successful reproduction. During the few days when copulation could result in conception, women's sexual interests often heighten (e.g., Hill, 1988; Regan, 1996; Roney & Simmons, 2013). Other competing motivations, such as appetite, diminish (e.g., Roney & Simmons, 2017; Gorczyca et al., 2016). Unlike most mammalian females, however, women have sex even during non-conceptive phases, which likely serves functions other than conception, such as strengthening the pair-bond and enhancing the likelihood a partner will invest in offspring (e.g., Strassman, 1981; Righetti et al., 2020; Gangestad et al., 2021). These non-conceptive periods are much more prevalent than the narrow conception window. It is therefore plausible that women have psychological adaptations that have specific functions during non-conceptive periods. These non-conceptive phases are not simply a “switching off” of the psychological responses to conceptive status. Behavior is a critical component of reproduction, so estrogens and progesterone

are involved not only in physiological changes, but also psychological adjustments that support goals relevant to both the conceptive and non-conceptive phases of the menstrual cycle.

Along these lines, Reynolds et al. (2018) investigated associations between women's progesterone levels across the cycle and their concerns about their levels of social support. Ancestrally, women's ability to successfully raise a child would have been enhanced by support from other people – their primary partners but also, perhaps, members of broader social networks of potential alloparents. Reynolds et al. (2018) proposed that, post-ovulation, and extending into early pregnancy, women should become attuned to and worry about the levels of social support they receive. Progesterone, which increases around the time of implantation and early pregnancy, may produce these effects. Two studies ($N = 100$ and 61) examining associations of progesterone levels with measures of anxiety in naturally ovulating women yielded qualified support for this proposal. In both studies, women's average progesterone across sessions positively covaried with their anxiety levels. In Study 2, within-woman variations in progesterone levels also predicted scores on a measure specifically tapping attachment anxiety (e.g., "I am afraid to lose my partner's love"). In Study 1, however, within-woman associations of progesterone levels covaried weakly and non-significantly with a single-item rating of anxiety level ("I feel anxious today"). In sum, these studies offered consistent evidence for an association between women's average progesterone levels and their anxiety (i.e., between-subjects effect). Evidence pertinent to whether women's level of anxiety tracks their progesterone levels, as they change across the cycle (i.e., within-subjects effect), was inconsistent. However, the measure of anxiety in the study was limited to a single item, and one not specifically targeted at women's worries about interpersonal support. This is insufficient to reject the hypothesis that progesterone drives within-cycle changes in anxiety about potential interpersonal support.

Reynolds et al. (2018) were careful to note limitations of their studies and findings. As they discussed, results could not speak to the direction of causality. Although progesterone may promote vigilance to levels of interpersonal support, it is also possible that stress-inducing experiences, perhaps especially ones that are interpersonal in nature, may lead to increased levels of progesterone (see Dinh et al., 2021). The latter possibility is perhaps most consistent with weak or variable associations of anxiety with within-woman progesterone levels. Progesterone secreted during stressful experiences is produced in the adrenal glands, where it is a biosynthetic precursor to cortisol (e.g., Herrera et al., 2016). Yet, the vast majority of within-woman variation is almost certainly of ovarian origin—progesterone levels massively increase post-ovulation.¹ Within-woman variations in progesterone produced during stressful experiences may be difficult to detect against this background of massive cyclic variation. Still, additional empirical work is needed before firm conclusions about true empirical patterns or their likely causal explanation can be drawn.

Hahn et al. (2020) performed a partial conceptual replication of Reynolds et al.'s work. They specifically examined hormonal associations across the cycles of naturally cycling women with anxious jealousy, a construct likely related to anxious attachment (sample item, "I worry that my partner might leave me for someone else"). They found no evidence for a within-woman association between progesterone levels and anxious jealousy (see also Stern et al., 2023, who found no within-woman association between progesterone levels and a general measure of jealousy; neither study reported associations of these outcome measures with women's average levels of progesterone). Though jealousy likely relates to concerns about partner involvement, social support, and more general anxiety, it pertains to a particular component of it.

¹ Arslan et al. (2023) found that one can predict women's progesterone levels based how many days before or after an LH surge a value is sampled with $r = .89$, which implies that, at a bare minimum, 80% [$.89^2$] of the variance in within-woman levels is uterine in origin.

Shifts in mood across the menstrual cycle have been of long-standing interest to researchers, especially in the context of premenstrual symptoms. The premenstrual phase (the last several days of the cycle) is not associated with the highest progesterone levels; indeed, during this phase, progesterone levels diminish. In any event, reviews of the literature offer no clear conclusion about how negative or positive mood shifts across phases (e.g., Romans et al., 2012; Stern et al., 2025). One recent study found a descriptively (and non-significant) negative within-woman association between progesterone and negative affect (Stern et al., 2025, Study 2). The study did not report associations with women's average progesterone levels.

In the current study, we further investigated associations between progesterone and women's anxieties about social support and their causes. In light of Reynolds et al.'s findings, we were interested in examining the robustness of both between-women (cycle average) and within-women associations of progesterone with worries about social support and negative emotionality. Though these associations are correlational in nature, their patterns can speak to underlying causal scenarios, to the extent that different causal scenarios expect different patterns of association (e.g., Grosz et al., 2020). Once again, if progesterone facilitates an adjustment in women's attunement to their social support, both between-women and within-woman associations are to be expected. If it exists, a robust association of average progesterone levels and worries about partner involvement or negative emotionality, even in the absence of within-woman associations, is an empirical observation that demands explanation. With the current sample, we were able to perform additional unique tests to probe the causal scenario at play. Associations were tested in a sample of 213 naturally ovulating women in heterosexual romantic relationships, of whom 180 participated in multiple lab sessions conducted across a month. Of the 180 women attending multiple lab sessions, 135 attended four sessions, 15 attended 3 sessions, and 30 attended 2 sessions, resulting in 645 total observations. During each session, women completed a questionnaire that included (a) a 9-item assessment of

women's positive and negative emotional states over the past two days (e.g., "distressed," "nervous," "happy," "sad"), and (b) a 3-item measure specifically targeting women's worries about their partners' levels of investment and involvement (e.g., "I worried about my partner's level of involvement in our relationship"). We assayed estradiol, progesterone, cortisol, and testosterone (or their metabolites) in urine collected the day of each session. We hence were able to assess women's average and within-cycle associations between hormone levels and psychological states, including both general negative affect and worries about partner investment specifically. In subsequent analyses, we examined whether effects held after accounting for individual differences, including neuroticism and perceived family support, as these individual differences could potentially drive associations between women's average progesterone levels and outcome variables. In addition, based on the results of initial analyses, we separately examined associations with mean progesterone levels during the follicular and luteal phases, as the source of the former is largely adrenal whereas the source of the latter is largely ovarian (e.g., Herrera et al., 2016). Especially as adrenal progesterone may be secreted in response to stress (Herrera et al., 2016), differential associations could help illuminate potential causal scenarios.

2. Methods

2.1 Sample and Preregistration

We enrolled 229 women in a four-session study on "Relationship dynamics and endocrinology." We advertised the study on the Department of Psychology's website for research participation and with flyers posted around the university's campus. Participants were required to be (a) aged 18-35, (b) naturally ovulating (not pregnant, breastfeeding, or using hormonal birth control), and (c) in a current heterosexual romantic relationship of at least three months duration. Of 229 recruited, 213 met requirements; 16 were excluded: 11 used hormonal contraceptives; one used a

contraceptive pill one week before the study; one learned she was pregnant after a second session; two were in non-heterosexual relationships; one was single.

As compensation, participants received credit toward a psychology course requirement or grade, and/or monetary compensation: \$25 for an initial 1.75-hour session and \$15 for each of three subsequent one-hour sessions. Participants received a \$20 bonus upon completing all sessions.

Participants were recruited to attend four lab sessions, but some attended fewer. The relationships of 11 participants (4.7%) ended during the study; we retained all sessions prior to the breakup. Two other participants claimed to not be with partners one session, only to resume the relationship by the next session. For them, we excluded the session the relationship was on hiatus. One participant attended multiple sessions but, through oversight, urine was collected only one session. For primary analyses examining within-woman variations in hormone levels, we analyzed data of the 180 women who had hormone assays for at least two qualifying sessions (mean number of sessions = 3.6).

Mean age was 20.9 years ($SD = 3.8$; range = 18-35 years). Mean partner age was 22.6 years ($SD = 5.9$; range = 17-53 years). Mean relationship length was 24.7 months ($SD = 29.6$; range = 3-228 months). Eighteen women were married (8.5%), 20 were unmarried and cohabiting with partners (9.4%), and 10 had children (4.7%). Most participants had one primary partner; two women (.9%) claimed to date a second person.

The data collection procedures and sample of participants used here were described in a preregistration for a larger project (see <https://osf.io/py5ke/>). The preregistered target sample size was 260. In March 2020, with 213 eligible participants, new enrollment ceased due to the COVID-19 pandemic. The 7 participants who had been enrolled at this time (and included in the 213) could not complete all four sessions. A number of hypotheses and relevant analyses were preregistered for the larger project, but the models we ran to assess the current set of associations were not

preregistered. Nonetheless, many elements of the models were preregistered (e.g., use of log-transformed hormone values, within-woman and cycle-average variance separation, random component selection). We retained these same elements in the current analyses. We note where analysis decisions have been guided by the preregistration (see <https://osf.io/py5ke/>). To be clear: the analyses done in the current study follow the logic of the analyses for the preregistered project; however, *we did not preregister the current hypotheses or specific analytic steps used in the current study*. As well, we note where we have run sensitivity tests for analyses to evaluate the robustness of results.

2.2 Procedures

2.2.1 Overview

A first session lasted ~1.75 hours. Sessions 2-3 lasted 45 minutes to 1 hour. Participants were administered multiple tasks each session, including questionnaires asking about their thoughts, feelings, desires, and behaviors over the past two days. Women also provided a urine sample at the lab, which was immediately frozen at -40°C. For sessions 2-4 (typically one session per week), women collected their first urination of the day at home and brought the sample into the lab, which was frozen at -40°C. For a description of the full procedures, see Dinh et al. (2022a), Supplemental Online Materials, S1.

2.2.2 Session Scheduling

On average, sessions were scheduled about 1 week apart. During Session 1, women reported the first day of their last menses and their typical cycle length, which were used to calculate current cycle day (where day 1 is the first day of menstruation). Based on these estimates, we targeted Session 2 for a peri-ovulatory day (depending on cycle length, typically day 10-15) or a mid-luteal day (typically day 19-25). If neither timeframe was upcoming, we targeted the next session for the early-to-mid follicular phase. Using similar guidelines (and updated information about last menses), we scheduled Sessions 3 and 4 at the conclusion of Sessions 2 and 3, respectively. Subsequent sessions

were scheduled to target the next cycle phase (e.g., mid-luteal following peri-ovulatory), accounting for cycle length and updated onset of menses. Supplementary Online Materials [SOM], S1, we list a table of all forward cycle days represented in the sample.

2.2.3 *Women's reports of mood and worries about partner involvement*

During each session, women reported their mood states from the previous two days. They also reported the extent to which they worried about their partner's investment in their relationship over the previous two days.

Negative and positive emotional states. Women were given (at each session) a list of nine “words that describe different feelings and emotions,” and asked to “Indicate the extent to which you have felt this way over the past two days” on a 1 (very slightly or not at all) to 7 (extremely) scale. The nine words were (a) distressed, (b) excited, (c) upset, (d) happy, (e) miserable, (f) sad, (g) nervous, (h) hopeless, and (i) joyful. Six descriptors were taken from the PANAS (Watson et al., 1988) to assess positive and negative emotions. We added three additional descriptors to target depressive emotion: miserable, sad, and hopeless. A factor analysis of responses to these items (with principal axis extraction and direct oblimin rotation) yielded two robust factors (first three eigenvalues = 4.03, 1.67, 0.76). One factor reflects negative emotionality; item loadings (pattern matrix) greater than .4 include: distressed, upset, miserable, sad, nervous, and hopeless. The second factor reflects positive emotionality; item loadings greater than .4 include: excited, happy, and joyful. SOM, S2, gives additional details. Because two factors emerged, we saved factor scores using tenBerge et al.'s (1999)'s method (which, compared to the regression method, more closely preserves pattern matrix covariation between factors); factor scores were correlated at $r = -.53$. In supplementary analyses, we separately constructed measures of negative affect, depressive affect, and positive affect by averaging responses to each of the three mood indicators within each set.

Worries about partner involvement. During each session, women responded to three items about the extent to which they worried about their partner's involvement and interest in their relationship over the past two days: (a) I felt insecure about my relationship; (b) I worried about my partner's level of involvement in our relationship; (c) I wanted to feel more intimacy from my partner. Participants responded on a 7-point scale, ranging from -3 (far less than usual) to +3 (far more than usual). A principal axis factor analysis of the three items yielded a single factor with item loadings of .82, .94, and .51, respectively. We saved factor scores (regression method) as a measure of this dimension. These factor scores correlated $r = .32$ with negative emotionality and $r = -.21$ with positive emotionality, indicating moderate overlap but substantial unique variance and thereby supporting our treatment of these measures as distinct outcome variables in our models.

Table 1 reports reliabilities (Cronbach α and, where possible, ω) for these outcome measures and all other measures and covariates in our analyses. Reliabilities (α) for outcome measures are good (e.g., .82 for negative emotionality) and adequate to excellent for covariates (.61 to .93).

Family support. We asked 13 questions at the first session about women's perceived family support (e.g., "There are family members I can depend on to help me if I really need it"; "There is no family member I can turn to for guidance in times of stress" (reverse-coded); see Cutrona & Russell, 1987). Participants responded on a 4-point scale ranging from 1 (strongly disagree) to 4 (strongly agree). A principal axis factor analysis strongly suggested a one factor solution (first three eigenvalues: 7.31, 1.04, 0.93). We saved factor scores (regression method) as a measure of this dimension.

Friend support. Participants responded to the same 13 items from the family support category with the wording modified to ask about friends (Cutrona & Russell, 1987). The same 4-point scale was used. A one factor solution fit best, and factor scores (regression method) were saved as a measure of this dimension.

Attachment Style. We administered the 17-item Adult Attachment Questionnaire (AAQ; Simpson et al., 1992) during the first session. Simpson et al. (1996) reported that two factors, interpretable as Avoidance and Ambivalence, sufficiently account for item covariance. We created scores for these two factors by averaging the items for each factor.

Personality. We used a short 30-item version of the NEO (McCrae & Costa, 2004); participants completed the inventory during the first session. Items were measured on a 7-point Likert scale from 1 (strongly disagree) to 7 (strongly agree); the five factors were scored by averaging the items for each factor.

2.3 Hormonal measures

2.3.1 Hormone assays

All assays were performed in the Comparative Human and Primate Physiology Center at the University of New Mexico with reagents and protocols were provided by the Clinical Endocrinology Laboratory at University of California, Davis. Estradiol and progesterone are excreted into urine in their metabolized form; thus, we analyzed for the major metabolites using antibodies for estrogen conjugates (E1C, capturing several metabolites) and pregnanediol-glucuronide (PdG). Intraassay coefficients of variation (CV), determined as the mean CV of duplicate determinations, were 2.5% for E1C and 4.4% for PdG. For E1C, interassay CVs were 14.1% for a high control and 18.5% for a low control. For PdG, interassay CVs were 10.4% for a high control and 12.9% for a low control. As E1C should reflect accumulated metabolites of estrogens, the dominant form of which is estradiol, we refer to E1C as estradiol (E). As PdG should reflect accumulated metabolites of progesterone, we refer to PdG as progesterone (P). Cortisol, which was assayed in its native form, yielded an interassay CV of 15.4% and intraassay CV of 4.5%.

Immunoreactive testosterone (T) was assayed following a deconjugation procedure that employed enzymatic hydrolysis using beta-glucuronidase (*Helix pomatia*, EMD Millipore #347420-

1MU with $\leq 2\%$ arylsulfatase activity, see Emery Thompson et al. 2012). For T, intraassay CV was 6.0% and interassay CVs were 13.5% for a low control and 11.4% for a high control.

All hormone levels were corrected for urinary specific gravity.

2.3.2 Log-transformation

Women's hormone levels are highly positively skewed. Some scholars argue that outcomes are affected by proportional changes in these hormones through most of their physiological range; thus, hormonal effects tend to be linear as a function of log-transformation (Jones, 1996; Jeyakumar et al., 2011). In separate analyses using the current data set and one other (Stern et al., 2021), log-transformed estradiol and progesterone levels predicted women's conception risk, estimated from day of an LH surge, better than raw levels did (Dinh et al., 2022a). Consistent with these findings and our preregistration, we used log-transformed estradiol, progesterone, testosterone, and cortisol levels in primary analyses. In robustness analyses, we verified the results using raw hormone levels.

2.3.3 Within-woman and woman-mean hormone levels

As specified in our preregistration, in primary analyses, we separately entered two components of each hormone level as predictors: the within-woman (woman-centered) component and women's cycle average (between-women) component. For each participant, we averaged logged estradiol, progesterone, testosterone, and cortisol levels across all sessions to compute cycle average values for each hormone. To compute within-woman changes for each hormone, we subtracted each woman's average hormone level from each of her session-specific logged levels. Within-woman hormone levels are of primary interest for hypotheses regarding cycle shifts. Although between-women hormone levels provide less relevant information for the causal hypothesis suggested, we included them as Reynolds et al. (2018) reported significant between-women associations and as discussed later, they can still help determine what process is occurring.

2.3.4 Mean follicular and luteal progesterone levels

In exploratory analyses, we examined associations with mean follicular and mean luteal progesterone levels separately. We categorized each woman's session as follicular or luteal in a stepwise fashion: (a) based on day relative to LH surge, (b) a backward count of days to next menses, and (c) forward count of days from last menses (see SOM, S3). If an LH surge was detected, we used a cutoff reported by Arslan et al. (2023) to assign a session as follicular (a day after the surge or earlier) or luteal (two days after the surge or later). If an LH surge was not detected, but a backward count of days prior to next menses was available, we assigned the session as follicular if backward count was 15 days or more prior to next menses and luteal if backward count was 14 days or fewer prior to next menses (Arslan et al., 2023). If a backward count was not available but a forward count of days from last menses was available, we assigned the session as follicular if forward count was 14 or fewer days after onset of last menses and luteal if 15 days or more after onset of last menses (Arslan et al., 2023). We then calculated each woman's average follicular phase progesterone level and luteal phase progesterone level.

2.3.5 Outlier treatment

Log-transformation typically reduces the number and influence of outliers. Thus, in primary analyses (which used log-transformed levels), we did not eliminate any outliers. In robustness analyses we eliminated outliers by two different criteria applied to the full dataset separately (see also Section 3.7 below). Outliers were determined by these criteria: values exceeding 3 times the interquartile range below the first quartile or above the third quartile (a standard criterion for extreme outliers; Tukey, 1977); and values falling more than 3 standard deviations from the mean (e.g., Jones et al., 2018a, b). For raw hormone levels, the first criterion removed 2.2% of estradiol, 2.6% of progesterone, 1.1% of cortisol and 1.1% of testosterone; similar to the first criterion, the latter removed 2.0% of estradiol, 1.7% of progesterone, 2.0% of cortisol, and 1.7% of testosterone. No log-transformed estradiol, progesterone, or cortisol values were outliers by the first criterion;

only a single testosterone value (0.2%) was identified as an outlier by the first criterion. The second criterion removed the following percentage of values: 1.1% of estradiol, 0.5% of progesterone, 0.2% of cortisol, and 0.5% of testosterone.

2.3.6 Luteinizing Hormone (LH) Surge Detection

For all sessions targeted for a peri-ovulatory phase, we gave women a set of Pregmate® LH strips to use on up to two days leading up to and four days following the session (sensitivity of 25 mIU/ml, which maximizes detection of ovulation; Leiva et al., 2017). We asked women to dip strips in urine on scheduled days, record results, and bring test strips to their next session for us to confirm. On potential peri-ovulatory days, we also applied LH tests to that session's urine sample. A positive LH surge was detected for 102 participants (57%) who completed at least two sessions. Reasons for not detecting surges were varied: participation for less than a full cycle; anovulatory cycles (perhaps 30% in young populations; e.g., Roney & Simmons, 2013); lack of follow-up assessments after Session 4; less than perfect compliance with instructions to use LH strips. For current analyses, we treated whether an LH surge was detected as a covariate in robustness analyses, as women who do not experience an LH surge during a cycle could experience lower and less variable hormone levels (e.g., Roney et al., 2013).

2.4 Statistical Analyses

Multilevel linear regression models, with observations nested within women, were conducted using the lme4 package on R version 4.0.2. Parameters were estimated via restricted maximum likelihood, with Satterthwaite approximation for degrees of freedom. As stated above, the following statistical models to test the hypotheses were *not* preregistered. Nonetheless, to minimize researcher degrees of freedom, we (1) followed as many aspects of preregistered analyses as possible (e.g., random effect component removal), (2) built statistical models that were very similar to prior research (Reynold et al., 2018), and (3) ran a wide variety of sensitivity analyses (e.g., raw hormone

values, outliers included/excluded) to assess the extent to which results were robust to small analysis decisions.

2.4.1 *Dependent Variables*

Negative emotionality. We ran multilevel models on negative emotionality, examining associations with woman-mean and within-woman components of each hormone's levels.

Positive emotionality. Based on past findings, our primary focus was on negative emotionality. As supplementary, exploratory analyses, we ran the same multilevel models substituting positive emotionality for negative emotionality.

Worries about partner involvement. We ran the same multilevel models on worries about partner involvement.

2.4.2 *Predictors*

Hormonal measures. Hormonal predictors included eight variables: within-woman estradiol, cycle average estradiol, within-woman progesterone, cycle average progesterone, within-woman testosterone, cycle average testosterone, within-woman cortisol, and cycle average cortisol.

Menses. While only 10% of all sessions occurred during menstruation, 34% of participants (60) contributed a menses session, which should yield sufficient within-woman variation to estimate the effect of menses. A dummy-coded menses variable (1=currently menstruating; 0=not currently menstruating) was included as a control variable, as menstruation could potentially affect mood independently of hormone levels.

Weekend effect. We initially ran analyses with an additional covariate: How many, of the last two days, fell on a weekend (Saturday or Sunday). As most individuals do not work or attend classes on weekends, participants could have potentially chosen to spend more time with their partners and engaged in less stressful activities over weekends. We coded it in two ways: first, we treated only sessions run on Monday as overlapping with the weekend; second, we treated Monday sessions as

overlapping with two weekend days and Tuesday sessions as overlapping with one weekend day. Neither variable predicted outcome measures or changed results reported in text. Hence, this covariate was dropped in primary analyses. Additional details are displayed in SOM, S4.

2.4.3 Random components

In all analyses, we modeled random intercept variation across participants. Which random slopes (if any) to include for final analysis was assessed in a series of model comparisons using backward elimination procedures described by Matuschek et al. (2017; see also Bates et al., 2015) to control for Type I error rates while retaining power to test fixed effects. We began with a random effects model that included random slope variation for all within-woman hormone components (with no covariance terms between random effects). We then ran a model with the random slope with the weakest effect removed. If the log-likelihood ratio chi-square test between models yielded $p \leq .2$, we retained all random slope terms in a final model. If $p > .2$, we dropped the weakest and then ran the model with the second weakest random slope removed and compared it to the model with the remaining two random slopes term retained. Again, at this step, if $p \leq .2$, we retained two random slope terms. If $p > .2$, we dropped this term and proceeded to test whether the one remaining random slope should be retained. If at least one random slope term was included, we examined whether covariances between random slope terms and the intercept had $p \leq .2$. We did not model random slope variation for menses, a control variable, as doing so may sacrifice power without affecting Type I error rates for fixed effects of interest (Barr, 2013).

R markdown files reporting output document selection of random slope terms in all analyses presented in text and in SOM; we summarize them in SOM, S5.

2.5 Data availability

Data, analysis scripts, and output for the study are available on Open Science Framework: https://osf.io/8tp23/?view_only=55ef88eff1a14b578ffc45afa9fe30bd

3. Results

3.1 Negative emotionality

Three between-women hormonal composites each positively predicted negative emotionality: cycle average progesterone, $\gamma = 0.23$, 95% *CI* [0.11 0.35], $t(179.98) = 3.69$, $p < .001$; cycle average testosterone, $\gamma = 0.15$, 95% *CI* [0.03 0.27], $t(177.7) = 2.31$, $p = .022$; cycle average estradiol, $\gamma = -0.14$, 95% *CI* [-0.26 -0.02], $t(181.1) = -2.22$, $p = .028$. Just one within-woman hormonal component predicted negative emotionality: within-woman changes in cortisol, $\gamma = 0.07$, 95% *CI* [0.00 0.15], $t(140.7) = 2.02$, $p = .045$. Effects of within-woman components of progesterone, testosterone, and estradiol were near-zero or weak; in two of three instances, effects—including of progesterone—descriptively ran in the direction opposite the effect of the woman-mean component. See Table 2 and SOM, S6.²

In separate analyses, we examined predictors of negative affect and depressive affect (i.e., averaged scores calculated from specific items rather than the factor composed of all items). Results were generally similar to each other, and similar to those for overall negative emotionality. Cycle average hormonal effects were slightly stronger for depressive affect. Details are displayed in SOM, S7.

3.2 Positive emotionality

We detected no hormonal associations with positive emotionality. For all cycle average levels, $p > .58$. For all within-woman hormonal changes, $p > .24$. Details are displayed in SOM, S8.

3.3 Worry about partner involvement

² Hormonal predictors were entered simultaneously to control for potential effects of multiple hormones. One can nonetheless ask whether within-woman progesterone levels predict negative emotionality without within-woman variations in other hormones entered as predictors. In fact, the within-woman effect of progesterone was virtually unchanged: $\gamma = 0.00$, $t(449.19) = -0.06$, $p = .952$. See SOM S6 for additional details.

The three cycle average hormonal predictors of negative emotionality also tended to positively predict women's worry about their partners' involvement: cycle average progesterone, $\gamma = 0.14$, 95% *CI* [0.02 0.26], $t(169.6) = 2.14$, $p = .033$; cycle average estradiol, $\gamma = -0.16$, 95% *CI* [0.02 0.30], $t(169.1) = -2.28$, $p = .024$; cycle average testosterone, $\gamma = 0.13$, 95% *CI* [0.00 0.26], $t(166.1) = 1.89$, $p = .061$, though the effect for testosterone fell short of statistical significance. The effect size for progesterone's effect on worry about partners was notably smaller than it was for negative emotionality. Our analyses detected no effects of within-woman hormonal changes, all $t \leq .32$, *ns*. See Table 2.³

3.4 Interactions between progesterone and estradiol levels

As progesterone and estradiol may have antagonistic effects, one may expect interactive effects of progesterone and estradiol. We added interactions between average and within-woman components of these hormones (average estradiol $\times$ average progesterone, within-woman estradiol $\times$ within-woman progesterone, average estradiol $\times$ within-woman progesterone, within-woman estradiol $\times$ average progesterone) as predictors of negative emotionality and worry about partner investment. We detected no interactive effects. SOM, S9, provides details.

3.5 Control for detection of LH surge

Potentially, cycle average hormone levels could be stand-ins for other variables varying across individuals or across cycles, where these other variables (or a subset of them) drive associations. In exploring this possibility, we first examined the effects of a detected LH surge. In a binomial regression analysis, we found that mean progesterone levels (odds ratio = 1.61, 95% *CI* [1.15, 2.30] $p = .007$) were positively associated with detection of an LH surge sometime during a woman's month of participation; mean estradiol, testosterone, and cortisol levels fell short of

³ One can ask whether within-woman progesterone levels predicted worries about partner investment without within-woman variations in other hormones controlled. As with negative emotionality, the effect remained virtually unchanged. $\gamma = 0.00$, $t(389.70) = -0.16$, $p = .870$.

significantly predicting this outcome (see SOM, S10). This association may very well be explained by the fact that anovulatory cycles (less often associated with a detectable LH surge) tend to be characterized by low luteal-phase progesterone levels, as the overwhelmingly dominant source of luteal-phase progesterone is the corpus luteum, which forms after ovulation (Fritz & Speroff, 2011).

We effect-coded whether an LH surge had been detected (-1 = no, 1 = yes) and entered it as a control variable. This variable did not predict either negative emotionality ($t[174.8] = -1.24, p = .218$) or worry about partner investment ($t[166.4] = -.36, ns$). Furthermore, inclusion of this control did not meaningfully alter effects of woman-mean hormonal predictors. See SOM, S10, for details. Thus, associations between women's average progesterone levels and these variables were likely not merely a function of whether ovulation occurred.

3.6 Control for participant age

We included participant age as a covariate in additional analyses because age might confound the relationship between hormone measures and outcome variables. Its inclusion did little to change hormonal associations. We report details in SOM, S11.

3.7 Individual differences

We examined associations between cycle average progesterone levels, controlling for cycle average levels of other hormones, with 9 individual difference measures. Stable individual differences may be confounders which give rise to state negative emotionality or worries, and simultaneously affect average progesterone levels. If so, we can ask, how well do average hormone levels predict negative emotionality or worries, independent of individual differences?

3.7.1 Hormonal associations with individual differences

First, in light of evidence that progesterone levels rise in response to low levels of social support (e.g., Wirth, 2011), we examined associations of cycle average progesterone with measures of perceived family support and friend support. Second, as romantic attachment styles relate to

perceptions of support from a partner, we examined associations of cycle average progesterone with two factors of the Attachment Scale: avoidant attachment and ambivalence attachment. Third, we examined associations of cycle average progesterone with measures of the Big Five personality traits. Results are reported in SOM, S12. Cycle average progesterone levels were associated negatively with perceived family support, $r = -.22, p = .004$. Perhaps relatedly, cycle average progesterone levels were also positively associated with neuroticism, $r = .16, p = .031$ and negatively associated with extraversion, $r = -.21, p = .004$. These three individual differences covaried among themselves, $r = .20$ to $-.31$, all $p \leq .007$. No other associations with individual differences were found to be significant.

3.7.2 Control for individual differences

We reran analyses examining cycle average and within-woman hormonal predictors of negative emotionality and worries about partner investment, with individual difference confounds entered as additional control variables. Details are given in SOM, S12. Independent of all other individual difference variables, only neuroticism significantly predicted negative emotionality and worries about partner, $p < .001$ and $p = .019$, respectively. The effect of cycle average progesterone levels on negative emotionality remained significant, $p = .017$, though controlling for individual differences reduced the effect size of cycle average progesterone on negative emotionality by about 30% (.15 vs. .22). The association between cycle average progesterone and worry about partner investment fell short of significant, $p = .062$. Cycle average estradiol levels remained associated with worry about partner investment, $p = .013$.

Naturally, proper interpretation of these associations with individual differences controlled depends on an accurate causal model (e.g., Wysocki et al., 2022). Do individual differences cause both hormone variation and worries/negative emotionality, such that the causal impact of hormones is best assessed with individual differences controlled? Or is a different causal scenario true? For

instance, might progesterone affect individual differences, which then predict worries and negative emotionality? In this case, controlling for individual differences partials out causal impacts of progesterone mediated through individual differences. We cannot say at this point. It does appear, however, that cycle average hormonal components have weaker effects when individual differences are controlled, a finding that may inform interpretation of between-women hormonal associations.

3.8. Separate prediction of follicular and luteal mean progesterone levels

In additional exploratory analyses, we examined whether cycle average progesterone effects on outcomes were driven more strongly by follicular phase or luteal phase levels. Progesterone levels during the follicular phase, prior to ovulation, predominantly reflect production by the adrenal glands. Adrenal progesterone is substantially converted into cortisol (Taraborrelli, 2015). Its levels increase in response to the cortisol stress response. By contrast, most of progesterone produced during the luteal phase emanates from the corpus luteum, which forms after ovulation (Taraborrelli, 2015). Luteal progesterone plays important roles in preparing the maternal endometrium for implantation and pregnancy, should conception occur.

In analyses examining hormonal predictors of negative emotionality, substituting cycle average progesterone level with either follicular or luteal average progesterone level, we found that both follicular and luteal phase average progesterone levels had significant predictive effects: follicular: $\gamma = 0.26$, 95% *CI* [0.06, 0.46], $t(157.1) = 2.61$, $p = .010$; luteal: $\gamma = 0.21$, 95% *CI* [0.04, 0.38], $t(150.1) = 2.42$, $p = .017$. These patterns suggest that the association between mean progesterone levels and negative emotionality is not solely a product of the luteal phase (i.e., post ovulation). See SOM, S13.

By contrast, in analyses examining hormonal predictors of worry about partner investment, but which substituted cycle average progesterone level with either follicular or luteal average progesterone level, we found that only luteal phase average progesterone levels had a significant

predictive effect: follicular: $\gamma = 0.05$, 95% *CI* [-0.15, 0.26], $t(153.8) = .50$, *ns*; $\gamma = 0.31$, 95% *CI* [0.14, 0.48], $t(143.7) = 3.57$, $p < .001$. See SOM, S13.

3.9 Robustness analyses examining sensitivity to outliers and effects of raw hormone levels

To assess the sensitivity of effects to possible outliers and log-transformation, we ran our primary analyses examining effects of hormone levels with outliers removed (SOM, S14) and using raw hormone levels (as opposed to log-transformed hormone levels; SOM, S15). All cycle average effects of progesterone, estradiol, and testosterone in negative emotionality and worry about partner investment reported above were also detected when outliers were removed from analyses using log-transformed hormone levels, with one exception: cycle average progesterone levels did not predict worry about partner investment when outliers more than 3 SD from the mean were removed. In analyses using raw hormone levels, without outlier removal, all cycle average hormonal effects on negative emotionality and worry about partner investment reported above again emerged, with one exception: cycle average estradiol levels did not significantly predict negative emotionality. See S9 for additional details on outlier removal.

4. Discussion

Previous research found associations between naturally cycling women's progesterone levels and their anxiety (Reynolds et al., 2018). In two studies ($N = 100, 61$), they found that women's cycle average progesterone levels positively predicted their anxiety. In one study, but not the other, they detected a positive association between within-woman changes in progesterone levels and anxiety. In the study that did not detect within-woman effects, anxiety was measured with a single self-report item. Furthermore, the measure of anxiety differed across studies; within-woman changes in progesterone levels were associated with attachment anxiety (e.g., fear of losing a romantic partner's love) specifically. Using a larger sample of romantically involved, naturally cycling women assessed multiple times over the course of a month ($N = 180$), the current study examined cycle

average and within-woman hormonal associations with a measure of general negative emotionality, as well as a measure specifically tapping worries about partner relationship involvement. We additionally simultaneously controlled for women's levels of estradiol, testosterone, and cortisol. Positive associations of cycle average progesterone levels with negative emotionality and worries about partner involvement strongly replicated. Our study detected no associations of within-woman progesterone levels with these outcomes. The latter effects were near-zero in our sample and, in fact, descriptively ran in negative directions, an initial indication that the original causal scenario might be inadequate. No associations between hormones and positive emotionality were found; given that positive emotionality is best represented as a distinct construct from negative emotionality (Watson & Tellegen, 1985), this result isn't surprising, or relevant for interpretation of other findings. On negative emotionality and worries about partner investment, there were negative and positive associations of cycle average estradiol and testosterone levels, respectively. That is, women who had cycles characterized by lower estradiol and higher testosterone tended to experience more negative affect and worry about their partners' investment.

Subsequent analyses showed that, when potentially confounding individual differences—perceived family support, extraversion, and neuroticism—were controlled, the magnitude of associations of cycle average hormone levels with negative affect were reduced.

A key question arises: What causal scenario(s) explains the patterns of associations observed? We consider two primary candidates: higher progesterone levels cause increases in anxiety levels; alternatively, increases in anxiety levels (or stressors giving rise to experienced anxiety) cause increases in progesterone levels.

4.1 Scenario 1: Progesterone affects anxiety

Studies on humans and non-human animals provide evidence that progesterone affects individuals' responses to threats, albeit in complex and, at times, contradictory ways (e.g., Bitran et

al., 1995; Reddy et al., 2005; van Wingen et al., 2008; Andréen et al., 2005; Bäckström et al., 2011; Conway et al., 2007; Li et al., 2022; Maner & Miller, 2014). However, the current findings challenge any straightforward causal account that posits progesterone's direct influence on negative emotionality or worries about partner involvement in naturally cycling women. Progesterone causing anxiety can explain positive cycle average effects. However, causal effects should also generate within-woman associations between progesterone and negative affect/partner concerns. This study detected no such effects—indeed, descriptively, these effects ran in negative directions, such that confidence intervals are consistent with only weak positive effects (see also Hahn et al., 2020; Stern et al., 2025, Study 2). And, though Reynolds et al. (2018) detected positive effects in one study, no such effects were detected in their other study. The mean effect, averaged across all studies, is positive but near-zero (for details, see SOM, S16).

We acknowledge potential reasons why the effects of within-woman variation in progesterone levels could be underestimated in our study. First, the rate at which progesterone is metabolized into allopregnanolone and pregnanolone (which themselves could have anxiolytic effects) leads the latter to peak 2-3 days after peak progesterone levels (see Sundström-Poramaa et al., 2020). Second, Reynolds et al. (2018) found associations of within-woman progesterone levels with a specific measure of attachment anxiety (sample item, “I am afraid to lose my partner’s love”). We used a measure with overlapping content, but our results can only indirectly speak to effects on Reynolds et al.’s (2018) specific measure.

As noted earlier, an additional conceptual replication of Reynolds et al. (2018) used anxious jealousy as an outcome variable and found no significant within-woman associations (Hahn et al., 2020). Hahn et al. (2020) and the current study used measures similar to Reynolds et al. (2018), but these results can only indirectly speak to the positive effects on Reynolds et al. (2018) specific measure of attachment anxiety. Nonetheless, the lack of within-woman changes in the present

analysis is consistent with a related body of research, which does not find consistent evidence for cycle changes in jealousy (see Stern et al., 2023 for a review and additional tests; with respect to women's intrasexual competitiveness, another potentially related domain of outcomes, recent results in large-scale studies have been inconsistent; see, e.g., Stern et al., 2023, 2025; Blake, 2022; Arthur et al., 2022).

4.2 Scenario 2: Stressors and threats affect progesterone levels

We turn to an alternative account: Rather than progesterone influencing anxiety levels, anxiety-evoking stressors and threats could affect progesterone levels. Indeed, in a variety of mammalian species, physical and social stressors lead to the adrenal production of both cortisol and progesterone (reviewed in Herrera et al., 2016). In naturally cycling women too, a physical stressor (a cold pressor test) or social stressor were found to lead to increases in progesterone (e.g., Herrera et al., 2016; Dinh et al., 2021), effects that may be mediated by hypothalamic release of ACTH (e.g., Genazzani et al., 1998; also Mann & Barraclough, 1973; Resko, 1969). The precise function of stress-dependent progesterone production is not fully understood. Possibilities include it being a reservoir for additional biosynthesis of cortisol, a negative feedback signal suppressing additional cortisol release, or having a specific role in social support seeking (e.g., Brown et al., 2009; Wirth, 2011; Maner & Miller, 2014).

If stress impacts progesterone release, and individuals differ in the extent to which they systematically experience negative emotionality, negative emotionality will be associated with their average levels of progesterone, as both the current study and past studies have found. Specific anxious experiences, varying in intensity across time, should be associated with within-woman changes in progesterone; hence, within-woman progesterone levels too should possess non-zero positive associations with negative affect and worries. However, anxiety affects *adrenal* production of progesterone. Ovarian production of progesterone (by the corpus luteum) post-ovulation massively

affects within-woman variation. The overwhelming majority of *within-woman* variation in progesterone levels, then, is variation in levels produced in the ovaries, unaffected by anxiety-provoking stressors (as noted earlier, estimated to be 80% of the variance or more; Arslan et al., 2023). Within-woman changes in ovarian progesterone levels across women's cycles may swamp adrenal progesterone responses to stressful experiences, leading within-woman effect sizes to be small and often undetected, even in reasonably large samples.

There exists one complication in our findings that challenges this interpretation for one measure, worries about partner investment. We conducted more direct tests of whether luteal – and, hence, ovarian – progesterone is especially associated with negative emotionality and worries. We separately examined associations with mean follicular and mean luteal progesterone levels. Mean follicular progesterone levels positively predicted negative emotionality, consistent with adrenal progesterone being associated with negative emotionality.⁴ By contrast, only luteal progesterone levels predicted worry about partner investment. If, as we suggest, stressful experiences result in progesterone production, then cycle average progesterone levels should predict negative emotionality and worries about partner investment in the follicular phase. Hence, these findings, though consistent with the interpretation that stressful experiences reflected in our measure of negative emotionality lead adrenally produced progesterone levels to rise, are not consistent with worries about partner investment elevating adrenally produced progesterone levels.

That luteal phase progesterone drove the effect of progesterone on worries about partner investment is consistent with Reynolds et al.'s (2018) interpretation that progesterone drives concerns about social support. Perhaps importantly, Reynolds et al. (2018) found a within-woman

⁴ As follicular progesterone is largely secreted by the adrenal gland, whereas the bulk of luteal progesterone is ovarian in origin, if stresses cause increases in adrenal progesterone, negative emotionality should covary more strongly with mean follicular than mean luteal progesterone levels. Descriptively, it did so in our analyses, though the difference in effect size is small.

effect of progesterone on concerns about social support, but not general anxiety. At the same time, the lack of a within-woman progesterone effect in the current data is inconsistent with this interpretation.

Overall, the contrasting results are hard to interpret in a causal framework. Possibly, within-woman effects were underestimated, and progesterone does reliably affect worries about partner investment. But we also cannot rule out the possibility that the contrast between effects of luteal and follicular progesterone levels on worries about partner investment is exaggerated in this sample, a reflection of sampling variability. Worries and negative emotionality were moderately correlated ($r = .32$). Worries may be a manifestation of a broader construct of negative emotionality. Indeed, the mean effect for worries about partner investment was considerably smaller than the effect for negative emotionality, consistent with the former having been driven by the latter—though, if that were the case, one should not expect different patterns of separate effects for follicular and luteal progesterone. The reasons for the overall pattern of results cannot be resolved by this single study; additional work is needed.

4.3 Associations with within-woman cortisol and average levels of cortisol, estradiol, and testosterone

The association between within-woman cortisol changes and negative emotionality is expected. Cortisol, exclusively produced in the adrenal glands, is expected to vary from day to day as a result of variation in the experience of metabolic demands and other stressors. At the same time, *average* cortisol levels were not associated with negative emotionality, a null result that further complicates findings because, as previously noted, stressors elicit the production of both cortisol and progesterone. Thus, an average progesterone effect without an average cortisol effect may have implications for whether the second causal scenario is adequate. More detailed physiological work is needed to contextualize this effect.

Analyses yielded associations of cycle average levels of both estradiol and testosterone with negative emotionality and worries about partner involvement. In these analyses, estradiol serves as a suppressor variable. A suppressor variable has little to no association with a predicted variable by itself (and, perhaps, no direct causal association with the predicted variable). When added to a prediction equation, however, it receives a non-zero weight because its variation overlaps with variation in another predictor variable that is associated with the predicted variable, where the overlapping variation is unrelated to the predicted variable (Cohen & Cohen, 1975). As the unique variance in the other predictor variable is, proportionally, more related to the predicted variable with overlapping variance controlled, inclusion of the suppressor variable increases the predictive ability of the other predictor variable. Consistent with this interpretation, when cycle average progesterone level was excluded as a predictor of negative emotionality, mean estradiol level had no significant effect, $t(185.6) = -.74$, *ns* (see SOM, S17). Cycle average estradiol levels are associated with cycle average progesterone levels, however, $r = .42$. Inclusion of cycle average estradiol levels thereby “removes” or “suppresses” variance in cycle average progesterone levels unrelated to negative emotionality. (We note that Reynolds et al. [2018; Study 1], found a similar pattern: a [nearly significant; $p = .066$] association of cycle average estradiol levels with anxiety when progesterone was a predictor as well, but no association of cycle average estradiol levels with anxiety without progesterone levels controlled.) A plausible interpretation is consistent with the second scenario we laid out above. Stressors do not appear to notably affect adrenal production of estrogens (e.g., Herrera et al., 2016). The association between cycle average progesterone and cycle average estradiol levels, therefore, are very likely due to correlated production of both hormones by the corpus luteum. Controlling for cycle average estradiol levels, then, should suppress (or remove) some variation in cycle average progesterone levels due to ovarian production, leaving proportionately

more variation due to differences in adrenal production—variation that is subject to effects of stressors.

Even with cycle average progesterone levels excluded as a predictor, cycle average testosterone levels positively predict negative emotionality (see SOM, S17). The independent association of cycle average testosterone levels, then, requires another explanation. Past work has shown both positive associations (e.g., Flores-Ramos et al., 2024) and negative associations (e.g., Pletzer et al., 2021) between women's testosterone levels and stress-reactivity. Future research may address the robustness of these associations as well as conditions that may moderate them.

4.4 Adaptations for early pregnancy?

As we discussed at the outset, Reynolds et al. (2018) proposed that associations between progesterone levels and anxiety may be driven by effects of progesterone on concerns about levels of support from important social partners, including romantic partners. These effects may reflect adaptations for very early pregnancy. Pregnancy is a vulnerable state and demands increased caloric consumption to support growth of a fetus. Early pregnancy thereby intensifies “women's need to generate a safe and supportive environment” (p. 38), which could potentially be achieved partly through increased worry and concern about sustainable levels of social support. Progesterone plays important roles in orchestrating physiological processes (e.g., intrauterine events) that support successful implantation of a fetus. It is reasonable to postulate that progesterone has also evolved to have psychological effects that support potential implantation of a conceptus and early pregnancy, should conception within a cycle occur. In a related vein, then, luteal-phase progesterone has been proposed to increase attentiveness to partner relationship involvement and, when levels are lagging, direct sexual interest toward primary partners to bolster partner involvement and commitment (e.g., Gangestad et al., 2021).

The current results challenge the proposal that elevated levels of progesterone during the luteal phase, when implantation may occur, adaptively increase general anxiety as well as worries specifically concerning levels of partner relationship involvement. Once again, this proposal expects that the massive increases in progesterone levels experienced during the luteal phase will bolster anxiety and concerns about partner support, yet the current study found little to no support for these links.

That said, the current results cannot speak to adaptations that involve progesterone's effects on psychological processes geared to regulate women's interpersonal relationships more generally. One recent study (using the same sample as Reynolds et al., 2018) found that women in the luteal phase, compared to follicular or peri-ovulatory phase, demonstrated an increased desire to affiliate with close others (Makhanova et al., 2025). Further, our results cannot adjudicate whether effects would be similar in a group of women experiencing the first few days or weeks of pregnancy. We hence reiterate Reynolds et al.'s (2018) call for future research to "explore whether rising progesterone levels correspond to specific behaviors that may help generate an optimal environment for pregnancy and motherhood, such as avoidance of unfamiliar or threatening men, risk aversion, support seeking, or relationship maintenance" (p. 38).

4.5 Limitations and Future Recommendations

We note a number of limitations of the current study and its ability to address causal questions. In that regard, we also add recommendations for future research.

4.5.1. Causal interpretation

The current study was motivated by a causal hypothesis: that progesterone, particularly during the luteal phase, increases concerns about social support and partner involvement. It should expect associations between both within-woman and average progesterone levels with measures of these outcomes and related outcomes (e.g., negative emotionality). Analyses found support for the

latter (replicating two studies reported by Reynolds et al., 2018) but not the former. Associations with average progesterone levels demand an explanation. Furthermore, whatever causal scenario generates these associations may further inform an understanding between hormonal variations and these and related constructs. We explicitly discuss a potential causal scenario to better inform future research (Grosz et al., 2020). It invokes causal phenomena that have been established in independent studies examining the impact of stress on adrenal hormone production (e.g., Herrera et al., 2016). We present analyses that allowed us to assess compatibility of this scenario with patterns of association. At the same time, we fully recognize that the correlational associations we present are limited in their ability to fully address causality of an average progesterone link with negative emotionality.

4.5.2 Potential issues with within-woman hormonal variation and its associations

We assessed hormone levels on days we measured outcome variables. Assays of urinary metabolites, however, may reflect serum hormone levels experienced a day or two prior to urine collection (e.g., Santoro et al., 2003). One might worry that the time-lag attenuates within-woman effects. Though we recognize this potential concern, there are reasons to think the time-lag plays a minimal role in explaining small within-woman associations. First, participants were asked to report their mood or emotional experiences over the past two days (the last 48 hours). This way of sampling mood means that, on average, reports are linked to mood 24 hours prior—which corresponds closely with the window captured by urinary hormone assays. Thus, while urinary concentrations may not reflect real-time hormonal states, they are aligned with the two-day retrospective time frame of the psychological measures we assessed. Second, estradiol and progesterone levels change some from day to day, but correlations between 1-day lags are very high. Hence, for instance, Arslan et al.’s (2023) imputed values (using backward count), based on serum levels across approximately 500 cycles in the well-curated BioCycle sample, give 1-day lag

correlations of .96 and .98 for estradiol and progesterone, respectively. Hence, even 1-day lagged values will have validity close to concurrent values; attenuation of correlations from “true” values should be very minimal, hardly sufficient to account for near-zero within-woman hormonal associations. (Indeed, in this same sample, associations between within-woman progesterone and sexual desires were found; Dinh et al., 2022a; 2022b.)

4.5.3 Potentially non-robust associations

Scholars have noted that many “significant” associations reported in journals have p -values that fall between .01 and .10. While seemingly suggestive of true effects, an issue arises: In well-powered studies, one should expect half or more of all p -values to fall *below* .01 (see, e.g., a blog by Rohrer, <https://www.the100.ci/2018/02/15/the-uncanny-mountain-p-values-between-01-and-10-are-still-a-problem/>). If so, one can be suspicious of strength of evidence for meaningful effects when a large majority of “significant” effects are not less than .01.

In the present study, it is worth distinguishing associations of average hormone levels with outcome variables that were targeted a priori and those emerging from inclusion of control variables. In our primary analyses, associations (with $p < .05$) for average progesterone, estradiol, and testosterone levels with negative emotionality and worries about partner involvement emerged. But only the associations with progesterone levels were targeted a priori, based on Reynolds et al.’s findings. For those effects, p -values were $<.001$ and .033—1 of 2 less than .01 (and, indeed, substantially less than .01). (Reynolds et al. reported p -values of .001 and .028 for these associations with anxiety and attachment anxiety in their two studies, respectively—again, 1 of 2 less than .01. In both Reynolds et al. and our study, the p -value for the association with negative emotionality was .001 or less.) By contrast, p -values for average estradiol and testosterone associations—which involved hormone levels included as control variables—ranged from .022 to .061. One can reasonably wonder whether these effects are, in fact, truly meaningful. For this reason, while these

associations could be real, we do not firmly conclude, based on the current study, that they truly are meaningful (see section 4.3 above). Additional work is needed.

4.5.4 Effect size interpretation

Associations with average progesterone levels were observed to be “statistically significant,” but are they of meaningful size? In our primary analyses, we standardized predictors and outcome measures, such that effect sizes are akin to standardized effects. For negative emotionality, the observed effect size was .23; for worries about partner involvement, it was .14. As these associations are between average progesterone levels with individual daily reports (rather than average response, where many time-varying factors may influence individual daily reports), we consider the former effect to be of meaningful size (see, for instance, Funder & Ozer, 2019, on the meaningfulness of small effect sizes in similar contexts). The latter is more modest (though potentially meaningful too; Funder & Ozer, 2019). Naturally, additional work may help pinpoint true effect sizes and their meaningfulness.

4.5.5 Recommendations for future research

Future studies can better test causal interpretations. Hence, for instance, non-experimental research that uses densely sampled, longitudinal designs that assess both hormone levels and psychological states daily (or more frequently), and with precise timestamping, would allow researchers to better establish temporal ordering and model time-lagged associations. While within-woman associations by design control for stable differences between woman, there is still potential confounding by time-varying covariates. Many variables may not change during short-term intensive longitudinal studies (e.g., relationship status), however, there are still time varying covariates that could drive associations (e.g., point in cycle, time since last seen partner). The impact of time varying covariates should be carefully considered. Studies that follow women across multiple cycles would also help determine what generates the consistent between-women association. If cycle average

effects are not driven by other individual differences, we would expect women to have more negative emotions during cycles they have higher average progesterone than their average cycle.

Results regarding associations with follicular and luteal phase average progesterone levels from the current study are provisional. We see these associations, however, as potentially important to assess potential causal scenarios. We recommend that future studies examine these associations with similar and related outcome measures. The study designs required for these analyses pose challenges, but we suggest that they may be needed to address important conceptual and empirical issues.

4.6 Summary

In a sample of 180 romantically involved, naturally cycling women followed over one menstrual cycle, cycle average progesterone levels predicted negative affectivity and concern over partner investment. Within-woman changes in progesterone levels did not reliably relate to variations in negative affectivity or worries about partner investment across the cycle. Overall, results may be most consistent with a scenario in which stressors systematically associated with negative affectivity generate elevations in circulating levels of progesterone of adrenal origin, rather than effects of progesterone on negative affectivity. Nonetheless, additional research is needed to further clarify causality.

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Declarations of conflicts of interest

None.

Author contributions

RTD: conceptualization; data curation; formal analysis; software; writing – original draft; writing – review and editing

TD: data curation; investigation; methodology; writing – review and editing

TAR: conceptualization; writing – review and editing

MET: conceptualization; data curation; funding acquisition; investigation; methodology; resources; writing – review and editing

SWG: conceptualization; data curation; formal analysis; funding acquisition; investigation; project administration; writing – original draft; writing – review and editing

Declaration of Generative-AI or AI-assisted technology in the writing process

No generative-AI or AI-assisted technology was used in the preparation of this manuscript.

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Table 1*Scale Reliabilities*

Scale / Subscale	α	ω_t	ω_h
Negative Emotionality	0.82	0.87	0.74
Worries About Partner Involvement	0.80		
Positive Emotionality	0.87		
Family support	0.93	0.95	0.78
Friend support	0.93	0.95	0.77
Extraversion	0.71	0.85	0.50
Agreeableness	0.61	0.77	0.62
Conscientiousness	0.81	0.89	0.59
Neuroticism	0.77	0.82	0.65
Openness	0.81	0.89	0.70
Avoidant Attachment	0.78	0.86	0.77
Ambivalent Attachment	0.66	0.76	0.35

Note. Because alpha has been shown to be a biased measure of reliability (see McNeish, 2018 for a non-technical review), omega analyses were conducted specifying three group factors to measure total reliability (ω_t) and reliability of a general factor (ω_h ; Zinbarg et al., 2005). For Worries About Partner Involvement and Positive Emotionality an omega analysis could not be conducted because the analysis requires six items to have sufficient degrees of freedom for model estimation.

Table 2

Results of multilevel models on negative emotionality and worry about partner involvement

Term	Negative Emotionality				Worry about Partner Involvement			
	Est.	SE	<i>t</i>	<i>p</i>	Est.	SE	<i>t</i>	<i>p</i>
Menses	0.05	0.11	0.43	.664	0.04	0.11	0.36	.716
Cycle average log-estradiol	-0.14	0.06	-2.22	.028	-0.16	0.07	-2.28	.024
Cycle average log-progesterone	0.23	0.06	3.69	<.001	0.14	0.06	2.14	.033
Cycle average log-testosterone	0.15	0.06	2.31	.022	0.13	0.07	1.89	.061
Cycle average log-cortisol	-0.06	0.06	-0.97	.331	-0.00	0.07	-0.01	.991
Within-woman log-estradiol	-0.00	0.03	-0.05	.964	0.01	0.04	0.29	.776
Within-woman log-progesterone	-0.01	0.03	-0.39	.693	-0.00	0.03	-0.13	.894
Within-woman log-testosterone	-0.02	0.03	-0.73	.464	-0.00	0.03	-0.09	.928
Within-woman log-cortisol	0.07	0.04	2.02	.045	-0.01	0.03	-0.28	.779

Note. All significant results ($p < .05$) are bolded, and all variables (excluding Menses but including outcome measures) are standardized. See additional details (e.g., df for all effects) in SOM, S6.